Git & Linux Placement MCQs with Answers

This document combines the original placement practice questions (Git and Linux MCQs) with their corresponding correct answers and explanations.

Part A — Git MCQs (Questions & Answers)

1. **Which command initializes a new Git repository in the current directory?**

    - *Answer:* **B. git init**
    - *Explanation:* `git init` creates a new `.git` directory and initializes the repository.

2. **Which command displays the current state of the working tree and staging area?**

    - *Answer:* **A. git status**
    - *Explanation:* `git status` shows modified, staged, untracked and other working-tree information.

3. **Which command stages a file for the next commit?**

    - *Answer:* **A. git add**
    - *Explanation:* `git add` places changes into the staging area (index).

4. **Which command creates a commit from the staged changes?**

    - *Answer:* **B. git commit**
    - *Explanation:* `git commit` records the staged snapshot in the local repository.

5. **Which command uploads local commits to a remote repository?**

    - *Answer:* **C. git push**
    - *Explanation:* `git push` transfers local commits/refs to a remote repository.

6. **Which command downloads remote changes and integrates them into the current branch?**

    - *Answer:* **A. git pull**
    - *Explanation:* `git pull` normally performs a fetch followed by integration, usually merge or rebase depending on configuration/options.

7. **Which command downloads remote references without integrating them into the current branch?**

    - *Answer:* **B. git fetch**
    - *Explanation:* `git fetch` updates remote-tracking references without changing the current branch.

8. **Which command creates a local copy of an existing remote repository?**

    - *Answer:* **B. git clone**
    - *Explanation:* `git clone` creates a new local repository based on an existing repository.

9. **Which command lists local branches?**

    - *Answer:* **A. git branch**
    - *Explanation:* `git branch` lists local branches by default.

10. **Which command creates and switches to a new branch in traditional Git syntax?**

    - *Answer:* **B. git checkout -b feature**
    - *Explanation:* `git checkout -b` creates the branch and checks it out. Modern Git also supports `git switch -c feature`.

11. **Which modern command switches to an existing branch?**

    - *Answer:* **A. git switch**
    - *Explanation:* `git switch` is designed specifically for switching branches.

12. **Which command combines one branch's history into the current branch using a merge commit when necessary?**

    - *Answer:* **A. git merge**
    - *Explanation:* `git merge` integrates another branch into the current branch.

13. **What is the purpose of git rebase?**

    - *Answer:* **B. Move/replay commits onto a new base**
    - *Explanation:* Rebase rewrites commit ancestry by replaying commits on top of another base commit.

14. **What is a fast-forward merge?**

- *Answer:* **B. A merge where the branch pointer can simply advance**
- *Explanation:* If the current branch is an ancestor of the branch being merged, Git can move the pointer forward without creating a merge commit.

15. **Which command shows commit history?**

    - *Answer:* **B. git log**
    - *Explanation:* `git log` displays commit history.

16. **Which command shows the differences between working-tree changes and the last commit/staged state as applicable?**

    - *Answer:* **A. git diff**
    - *Explanation:* `git diff` is used to inspect differences; without options it generally compares working tree with the index.

17. **Which command shows changes that are staged for commit?**

    - *Answer:* **A. git diff --cached**
    - *Explanation:* `git diff --cached` (or `--staged`) compares the index with HEAD.

18. **Which command removes a file from the staging area while keeping the file's working-tree changes?**

    - *Answer:* **A. git reset HEAD **
    - *Explanation:* `git reset HEAD <file>` (or `git restore --staged <file>`) unstages the file without normally discarding its working-tree content.

19. **Which command removes untracked files from the working tree?**

    - *Answer:* **A. git clean**
    - *Explanation:* `git clean` removes untracked files/directories according to its options. It can be destructive.

20. **Which command creates a lightweight or annotated reference to a specific point in history?**

    - *Answer:* **A. git tag**
    - *Explanation:* `git tag` creates tags, commonly used to mark releases.

21. **What is HEAD in Git?**

- *Answer:* **B. The current checked-out commit/reference**
- *Explanation:* HEAD identifies the currently checked-out commit, normally through the current branch reference.

22. **What is the staging area also called?**

- *Answer:* **A. Index**
- *Explanation:* Git's staging area is commonly called the index.

23. **Which area contains files as they exist in the currently checked-out version?**

- *Answer:* **A. Working tree**
- *Explanation:* The working tree is the set of checked-out files you edit.

24. **Which command temporarily stores uncommitted changes?**

- *Answer:* **A. git stash**
- *Explanation:* `git stash` saves local modifications and returns the working tree to a cleaner state.

25. **Which command lists stashed entries?**

- *Answer:* **A. git stash list**
- *Explanation:* `git stash list` displays the stash stack.

26. **Which command applies the latest stash and removes it from the stash list?**

- *Answer:* **A. git stash pop**
- *Explanation:* `git stash pop` applies the stash and removes it if successful. `git stash apply` keeps it.

27. **What does git revert do?**

- *Answer:* **B. Creates a new commit that reverses an earlier commit**
- *Explanation:* `git revert` is history-preserving: it creates a new commit that undoes the effect of a previous commit.

28. **What does git reset --hard primarily do?**

- *Answer:* **B. Moves HEAD and resets index and working tree to the target**
- *Explanation:* It resets HEAD, the index and working tree to the specified commit, potentially discarding uncommitted work.

29. **Which reset mode changes HEAD and the index but generally keeps working-tree changes?**

    - *Answer:* **B. --mixed**
    - *Explanation:* `git reset --mixed` (the default) moves HEAD and resets the index while keeping working-tree changes.

30. **Which reset mode moves HEAD but leaves both index and working tree unchanged?**

    - *Answer:* **A. --soft**
    - *Explanation:* `git reset --soft` changes HEAD only, leaving staged and working-tree content intact.

31. **What is a merge conflict?**

    - *Answer:* **B. A situation where Git cannot automatically reconcile changes**
    - *Explanation:* Conflicts occur when Git cannot automatically determine how overlapping changes should be combined.

32. **Which command helps identify the commit that introduced a particular line?**

    - *Answer:* **A. git blame**
    - *Explanation:* `git blame` annotates lines with the commit and author information associated with their latest change.

33. **Which command shows information about a particular commit?**

    - *Answer:* **A. git show**
    - *Explanation:* `git show` displays commit metadata and associated changes.

34. **What is a remote-tracking branch such as origin/main?**

    - *Answer:* **B. A local reference tracking a branch on a remote**
    - *Explanation:* `origin/main` is a remote-tracking reference representing the last known state of main on remote origin.

35. **What does origin usually represent?**

    - *Answer:* **A. The default remote name**
    - *Explanation:* `origin` is the conventional default name assigned to the remote from which a repository was cloned.

36. **Which command lists configured remotes and their URLs?**

    - *Answer:* **A. git remote -v**
    - *Explanation:* `git remote -v` shows remote names and fetch/push URLs.

37. **What is the main purpose of .gitignore?**

    - *Answer:* **B. Specify untracked files/patterns Git should normally ignore**
    - *Explanation:* A `.gitignore` file defines patterns for files Git should not normally consider as untracked.

38. **If a file is already tracked, adding it to .gitignore will automatically stop tracking it. True or False?**

    - *Answer:* **B. False**
    - *Explanation:* Gitignore does not untrack an already tracked file; you must remove it from the index, e.g., via `git rm --cached`.

39. **Which command creates a new local branch at the current commit without switching to it?**

    - *Answer:* **A. git branch feature**
    - *Explanation:* `git branch feature` creates the branch pointer but leaves the current branch checked out.

40. **Which command deletes a local branch after it has been merged?**

    - *Answer:* **A. git branch -d feature**
    - *Explanation:* `git branch -d` safely deletes a local branch when Git considers it merged.

41. **Which command displays the configured Git user name?**

    - *Answer:* **B. git config user.name**
    - *Explanation:* `git config user.name` reads the configured user.name value.

42. **Which command creates a new commit even if there are no changes, when allowed?**

    - *Answer:* **A. git commit --allow-empty**
    - *Explanation:* `git commit --allow-empty` creates an empty commit.

43. **What is the main purpose of a Git remote?**

    - *Answer:* **A. To provide a reference/location for another repository**
    - *Explanation:* A remote stores the connection information for another Git repository.

44. **Which statement about Git is correct?**

    - *Answer:* **B. Git is a distributed version-control system**
    - *Explanation:* Git is distributed: repositories contain substantial history and most operations can be performed locally.

45. **Which command can create an annotated tag?**

    - *Answer:* **A. git tag -a v1.0 -m "release"**
    - *Explanation:* `git tag -a` creates an annotated tag; `-m` supplies its message.

46. **Which operation rewrites existing commit history?**

    - *Answer:* **A. git rebase**
    - *Explanation:* Rebase rewrites commit ancestry. Other operations listed do not normally rewrite existing commits.

47. **Why should force-pushing rewritten history to a shared branch be avoided?**

    - *Answer:* **B. It can discard or invalidate collaborators' published history**
    - *Explanation:* Rewriting shared history can make collaborators' local histories diverge and may cause commits to become difficult to recover.

48. **Which command is commonly used to find a commit by searching commit messages?**

    - *Answer:* **A. git log --grep**
    - *Explanation:* `git log --grep='pattern'` filters commits by commit-message text.

49. **Which command shows the branches containing a particular commit?**

    - *Answer:* **A. git branch --contains **
    - *Explanation:* `git branch --contains` lists branches whose history contains the specified commit.

50. **What is a pull request primarily used for?**

- *Answer:* **A. Requesting review and integration of proposed changes**
- *Explanation:* A pull request is a collaboration mechanism on hosting platforms for reviewing and merging changes.

51. **Which is the best description of GitHub/GitLab/Bitbucket relative to Git?**

    - *Answer:* **B. They are hosting/collaboration platforms that can use Git repositories**
    - *Explanation:* These platforms provide hosting, collaboration, code review, CI/CD and other services around Git repositories.

Part B — Linux MCQs (Questions & Answers)

1. **Which command prints the current working directory?**

    - *Answer:* **A. pwd**
    - *Explanation:* `pwd` means print working directory.

2. **Which command lists files in a directory?**

    - *Answer:* **A. ls**
    - *Explanation:* `ls` lists directory contents.

3. **Which option commonly shows hidden files with ls?**

    - *Answer:* **B. -a**
    - *Explanation:* `ls -a` includes entries whose names begin with a dot.

4. **Which command changes the current directory?**

    - *Answer:* **A. cd**
    - *Explanation:* `cd` changes the shell's current working directory.

5. **Which command creates a new directory?**

    - *Answer:* **A. mkdir**
    - *Explanation:* `mkdir` creates directories.

6. **Which command creates an empty file if it does not exist and updates its timestamp if it does?**

- *Answer:* **A. touch**
- *Explanation:* `touch` creates an empty file or updates timestamps.

7. **Which command copies files?**

   - *Answer:* **A. cp**
   - *Explanation:* `cp` copies files/directories according to its options.

8. **Which command moves or renames a file?**

   - *Answer:* **A. mv**
   - *Explanation:* `mv` is used for moving and renaming.

9. **Which command removes a file?**

   - *Answer:* **A. rm**
   - *Explanation:* `rm` removes directory entries/files.

10. **Which option is commonly required to remove a directory recursively with rm?**

    - *Answer:* **A. -r**
    - *Explanation:* `rm -r` recursively removes directories and their contents.

11. **Which command displays the contents of a text file directly?**

    - *Answer:* **A. cat**
    - *Explanation:* `cat` writes file contents to standard output.

12. **Which command is commonly used to view a long file one screen/page at a time?**

    - *Answer:* **A. less**
    - *Explanation:* `less` provides interactive paged viewing and is commonly used for long files.

13. **Which command searches for matching text patterns in files?**

    - *Answer:* **A. grep**
    - *Explanation:* `grep` searches input for lines matching a pattern.

14. **Which command finds files/directories based on criteria such as name?**

    - *Answer:* **A. find**

- *Explanation:* `find` recursively searches directory trees using criteria such as `-name`, `-type` and `-size`.

15. **Which command displays running processes in a snapshot?**

    - *Answer:* **A. ps**
    - *Explanation:* `ps` reports current processes.

16. **Which command provides an interactive real-time process view on many Linux systems?**

    - *Answer:* **A. top**
    - *Explanation:* `top` displays processes and resource usage interactively.

17. **Which command sends a signal to a process?**

    - *Answer:* **A. kill**
    - *Explanation:* `kill` sends a signal to a process by PID; despite its name, it can send signals other than SIGKILL.

18. **What is the purpose of PID?**

    - *Answer:* **A. Identify a process**
    - *Explanation:* PID means process ID and uniquely identifies a process at a given time.

19. **Which command changes file permissions?**

    - *Answer:* **A. chmod**
    - *Explanation:* `chmod` changes permission bits.

20. **Which command changes file ownership?**

    - *Answer:* **A. chown**
    - *Explanation:* `chown` changes the user and/or group ownership of files.

21. **Which command changes group ownership?**

    - *Answer:* **A. chgrp**
    - *Explanation:* `chgrp` changes the group owner of a file.

22. **In rwx permissions, what does r represent?**

- *Answer:* **A. Read**
- *Explanation:* `r` is read permission.

23. **In rwx permissions, what does w represent?**

    - *Answer:* **A. Write**
    - *Explanation:* `w` is write permission.

24. **In rwx permissions, what does x represent?**

    - *Answer:* **A. Execute**
    - *Explanation:* `x` is execute permission for files and traversal permission for directories.

25. **What does chmod 755 file commonly mean?**

    - *Answer:* **A. Owner rwx; group r-x; others r-x**
    - *Explanation:* 7=rwx, 5=r-x, 5=r-x.

26. **What does chmod 644 file commonly mean?**

    - *Answer:* **A. Owner rw-; group r--; others r--**
    - *Explanation:* 6=rw-, 4=r--, 4=r--.

27. **Which symbol represents the root directory?**

    - *Answer:* **A. /**
    - *Explanation:* The filesystem root is `/`.

28. **Which symbol usually represents the current user's home directory in the shell?**

    - *Answer:* **A. ~**
    - *Explanation:* The tilde expands to the current user's home directory in typical shells.

29. **Which path represents the parent directory?**

    - *Answer:* **A. ..**
    - *Explanation:* `..` refers to the parent directory.

30. **Which path represents the current directory?**

    - *Answer:* **A. .**

- *Explanation:* `.` refers to the current directory.

31. **What is an absolute path?**

    - *Answer:* **A. A path beginning from the filesystem root or another defined root**
    - *Explanation:* An absolute path identifies a location independently of the current working directory, e.g., `/home/user/file`.

32. **What is a relative path?**

    - *Answer:* **A. A path interpreted relative to the current working directory**
    - *Explanation:* Relative paths are interpreted from the current working directory.

33. **Which command displays disk-space usage of mounted filesystems?**

    - *Answer:* **A. df**
    - *Explanation:* `df` reports filesystem disk-space usage.

34. **Which command estimates disk usage of files/directories?**

    - *Answer:* **A. du**
    - *Explanation:* `du` reports estimated space used by files/directories.

35. **Which command displays memory usage?**

    - *Answer:* **A. free**
    - *Explanation:* `free` reports free and used memory and related statistics.

36. **Which command displays the last lines of a file?**

    - *Answer:* **A. tail**
    - *Explanation:* `tail` prints the end of files.

37. **Which command displays the first lines of a file?**

    - *Answer:* **A. head**
    - *Explanation:* `head` prints the beginning of files.

38. **Which command is commonly used to follow a growing log file in real time?**

    - *Answer:* **A. tail -f**
    - *Explanation:* `tail -f` follows appended data as a file grows.

39. **Which operator redirects standard output to a file and overwrites it?**

    - *Answer:* **A. >**
    - *Explanation:* `>` redirects stdout and truncates/overwrites the destination file unless shell-specific conditions intervene.

40. **Which operator appends standard output to a file?**

    - *Answer:* **A. >>**
    - *Explanation:* `>>` appends stdout to the destination file.

41. **Which operator connects stdout of one command to stdin of another?**

    - *Answer:* **A. |**
    - *Explanation:* The pipe operator sends standard output to the next command's standard input.

42. **Which file descriptor is standard input?**

    - *Answer:* **A. 0**
    - *Explanation:* File descriptor 0 is stdin; 1 is stdout and 2 is stderr.

43. **Which file descriptor is standard output?**

    - *Answer:* **A. 1**
    - *Explanation:* File descriptor 1 is stdout.

44. **Which file descriptor is standard error?**

    - *Answer:* **A. 2**
    - *Explanation:* File descriptor 2 is stderr.

45. **Which command prints environment variables?**

    - *Answer:* **A. env**
    - *Explanation:* `env` prints the environment; `printenv` is another common command for this purpose.

46. **Which command displays a command's executable location when found in PATH?**

    - *Answer:* **A. which**
    - *Explanation:* `which` commonly reports the path of an executable resolved through PATH.

47. **Which variable stores the executable search path in a typical shell?**

    - *Answer:* **A. PATH**
    - *Explanation:* PATH contains directories searched for executable commands.

48. **Which command shows the exit status of the most recently executed foreground command in many shells?**

    - *Answer:* **A. echo $?**
    - *Explanation:* `$?` expands to the previous command's exit status in common shells.

49. **What does an exit status of 0 conventionally mean?**

    - *Answer:* **A. Success**
    - *Explanation:* Unix/Linux commands conventionally return 0 for success and a nonzero value for failure or other conditions.

50. **Which command creates a symbolic link?**

    - *Answer:* **A. ln -s**
    - *Explanation:* `ln -s` creates a symbolic (soft) link.

51. **Which command shows information about a command, including whether it is a shell builtin/alias/function/executable in many shells?**

    - *Answer:* **A. type**
    - *Explanation:* `type` is a shell builtin in common shells and identifies how a command name is resolved.

52. **Which command changes the current user's password?**

    - *Answer:* **A. passwd**
    - *Explanation:* `passwd` is used to change passwords, subject to system policy.

53. **Which command runs a command with another user's privileges when configured appropriately?**

    - *Answer:* **A. sudo**
    - *Explanation:* `sudo` executes commands with privileges authorized by sudo policy.